IYMC Qualification Round 2026

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Problem A

Problem Statement: Find all real roots of the function:

$$g(x) = (x-1) \cdot (x+4) \cdot (\ln x).$$

Solution

Before setting the expression to zero, we must determine the natural domain of the function.

1. **Domain of the function:** The natural logarithm $\ln(x)$ is only defined for strictly positive real numbers, so we require:
$$x > 0.$$
Thus, the domain of g is $\mathcal{D} = (0, +\infty)$. Any root must belong to this interval.

2. **Finding the roots:** A product equals zero if and only if at least one of its factors is zero:
$$(x-1) = 0 \quad \text{or} \quad (x+4) = 0 \quad \text{or} \quad \ln(x) = 0.$$
Let us check each factor:
   - $x - 1 = 0 \implies x = 1$. Since $1 > 0$, this is a valid real root.
   - $x + 4 = 0 \implies x = -4$. However, $-4 \leq 0$, which is outside the domain of $\ln(x)$ over the real numbers. Hence, -4 is not a valid root.
   - $\ln(x) = 0 \implies x = e^0 = 1$. This gives the same root we already found.

Therefore, the function has only one real root:

$$\boxed{x = 1}$$

Problem B

Problem Statement: Find all integers n such that this equation holds true:

$$2^n + 2^{-n} = \frac{5}{2}.$$

Solution

Let's make a substitution to simplify the equation. Let $u = 2^n$. Because $2^n > 0$ for any real number n, we know $u > 0$. Notice that $2^{-n} = \frac{1}{2^n} = \frac{1}{u}$.

The given equation becomes:

$$u + \frac{1}{u} = \frac{5}{2}.$$

Multiplying both sides by $2u$ to clear the denominators:

$$2u^2 + 2 = 5u \iff 2u^2 - 5u + 2 = 0.$$

We can factor this quadratic equation:

$$(2u - 1)(u - 2) = 0.$$

This gives two positive values for u:

$$u = 2 \quad \text{or} \quad u = \frac{1}{2}.$$

Now, let's substitute back $u = 2^n$:

- If $2^n = 2 \implies 2^n = 2^1 \implies n = 1$.
- If $2^n = \frac{1}{2} \implies 2^n = 2^{-1} \implies n = -1$.

Both 1 and -1 are integers, which satisfies the condition $n \in \mathbb{Z}$.
Thus, the integer solutions are:

$$\boxed{n = 1 \quad \text{and} \quad n = -1}$$

Problem C

Problem Statement: Determine the numerical value of the following expression without the use of a calculator:

$$E = \frac{\log_4(64) - (-1)^{2026}}{2} + \sum_{k=1}^{4} \frac{\cos\left(\frac{\pi}{2} + k\pi\right)}{(-1)^{k+1}} + \sqrt{\left(\frac{3}{\sqrt{3}}\right)^2} - 1.$$

Solution

Let us evaluate each part of the expression step by step.

1. **First term:**

$$\frac{\log_4(64) - (-1)^{2026}}{2}$$

Since $4^3 = 64$, we have $\log_4(64) = 3$. Also, since 2026 is an even number, $(-1)^{2026} = 1$. Therefore:

$$\frac{3 - 1}{2} = \frac{2}{2} = 1.$$

2. **Second term (the summation):**

$$\sum_{k=1}^{4} \frac{\cos\left(\frac{\pi}{2} + k\pi\right)}{(-1)^{k+1}}$$

Using the trigonometric identity $\cos\left(\frac{\pi}{2} + \theta\right) = -\sin(\theta)$, we get:

$$\cos\left(\frac{\pi}{2} + k\pi\right) = -\sin(k\pi).$$

Since $\sin(k\pi) = 0$ for every integer k, each numerator in the sum is exactly 0. The denominator $(-1)^{k+1}$ is either 1 or -1 (never zero), so every term in the sum is 0:

$$\sum_{k=1}^{4} 0 = 0.$$

3. **Third term:**

$$\sqrt{\left(\frac{3}{\sqrt{3}}\right)^2} - 1$$

Simplifying inside the square:

$$\frac{3}{\sqrt{3}} = \frac{3\sqrt{3}}{3} = \sqrt{3}.$$

Since $\sqrt{3} > 0$, taking the square and then the square root leaves $\sqrt{3}$:

$$\sqrt{(\sqrt{3})^2} - 1 = \sqrt{3} - 1.$$

Now we add all three parts together:

$$E = 1 + 0 + (\sqrt{3} - 1) = \sqrt{3}.$$

The exact value of the expression is:

$$\boxed{\sqrt{3}}$$

Problem D

Problem Statement: Prove that for every positive integer n,

$$S_n = 1 + 2026 + 2026^2 + \cdots + 2026^{n-1}$$

is divisible by 2027 whenever n is a multiple of 2026.

Proof

Let n be a multiple of 2026. This means $n = 2026 \cdot m$ for some positive integer $m \geq 1$.

Notice that 2026 is an even number ($2026 = 2 \times 1013$). Therefore, $n = 2 \times (1013m)$ is necessarily an **even integer**.

Now, let us examine the base 2026 modulo 2027:

$$2026 = 2027 - 1 \equiv -1 \pmod{2027}.$$

Using this congruence, every power 2026^k satisfies:

$$2026^k \equiv (-1)^k \pmod{2027}.$$

We can now reduce the entire sum S_n modulo 2027:

$$S_n = \sum_{k=0}^{n-1} 2026^k \equiv \sum_{k=0}^{n-1} (-1)^k \pmod{2027}.$$

Writing out this alternating sum:

$$\sum_{k=0}^{n-1} (-1)^k = 1 - 1 + 1 - 1 + \cdots + (-1)^{n-1}.$$

Because n is an even integer, the total number of terms in this sum is even (n terms). We can group every consecutive pair together:

$$\sum_{k=0}^{n-1} (-1)^k = \underbrace{(1-1) + (1-1) + \cdots + (1-1)}_{\frac{n}{2} \text{ pairs}} = 0.$$

Consequently:

$$S_n \equiv 0 \pmod{2027}.$$

This directly proves that S_n is divisible by 2027 whenever n is a multiple of 2026. □

Problem E

Problem Statement: A right triangle has legs of lengths a and b. A circle is inscribed in the triangle. Find the area of the region inside the triangle but outside the circle.

Solution

Let c denote the length of the hypotenuse of the right triangle. By the Pythagorean theorem:

$$c = \sqrt{a^2 + b^2}.$$

The desired area, A_{shaded}, is the area of the right triangle minus the area of the inscribed circle:

$$A_{\text{shaded}} = A_{\triangle} - A_{\bigcirc}.$$

1. **Area of the triangle:** Since the two legs are perpendicular, the area is simply:

$$A_{\triangle} = \frac{1}{2}ab.$$

2. **Radius of the inscribed circle (r):** Let r be the inradius. The tangents drawn from the right-angled vertex to the circle form a square of side length r. The remaining tangent segments from the other two vertices to the hypotenuse have lengths $a - r$ and $b - r$.

   Since the hypotenuse c is the sum of these two tangent segments:

$$(a - r) + (b - r) = c \iff a + b - 2r = c.$$

   Solving for r:

$$r = \frac{a + b - c}{2} = \frac{a + b - \sqrt{a^2 + b^2}}{2}.$$

3. **Area of the inscribed circle:**

$$A_{\bigcirc} = \pi r^2 = \pi \left(\frac{a + b - \sqrt{a^2 + b^2}}{2} \right)^2 = \frac{\pi}{4} \left(a + b - \sqrt{a^2 + b^2} \right)^2.$$

Subtracting the circle's area from the triangle's area, we obtain:

$$A_{\text{shaded}} = \frac{1}{2}ab - \frac{\pi}{4} \left(a + b - \sqrt{a^2 + b^2} \right)^2.$$

The area of the region inside the triangle but outside the circle is:

$$\boxed{A = \frac{ab}{2} - \frac{\pi}{4} \left(a + b - \sqrt{a^2 + b^2} \right)^2}$$